

3.6 Using Angle Relationships, Postulates and Theorems

Lesson Introduction

 Benchmark	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.		 Learning Target	I can solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.
 Benchmark Coherence	Building On MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.		Working Toward N/A	
 Essential Questions	How can you use the angle relationships, postulates, and theorems to solve problems?			
 Lesson Narrative	This lesson focuses on providing students practice in applying what they have learned about angle relationships to solve problems. The lesson features a Card Sort activity where students collaborate to match images with the name of angle pairs identified. Students will also match an answer card by using the angle pair relationships to create an equation and solve for x . Students then practice applying these relationships to solve problems independently.			
 Component Summary	3.6.1 Warm-Up (~5 min.)	Would You Rather with two ramps (<i>SE p. 154</i>)	3.6.3 Your Turn! (15 min.)	Practice using angle pair relationships to solve problems (<i>SE p. 156</i>)
	3.6.2 Activity (20 min.)	Card Sort using angle pair relationships to solve for variables (<i>SE p. 155</i>)	3.6.4 Wrap-Up (~5 min.)	Ticket Out the Door using angle pair relationships to find missing angle measures (<i>SE p. 157</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Congruent - Supplementary - Parallel Lines - Transversal		Blackline Master	For 3.6.2 Activity, teachers will need to make copies and cut out columns and rows of the blackline master to create sets to give to pairs or small groups of students.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may use the incorrect angle pair relationships when creating equations to solve. - Students may make errors related to solving equations. - Students may confuse when they are solving for a variable versus a missing angle.	Consider using information from the previous lessons to form a small group of students who are struggling working with angle relationships or solving equations. Facilitate these students as they work through 3.6.2 Activity.
	Literacy and Language Routines	Mathematic Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Group Presentations For the 3.6.2 Activity, consider having a different pair of students present their answers to the class for each matching set. Students should use proper academic language when describing the pairings and when justifying how to create the equations. After a pair finishes, have another student restate why the students made the matches they did.	MA.K12.MTR.5.1: Patterns and Structure Students will use their knowledge of the angle pair relationships formed by two parallel lines and a transversal, linear pairs, and vertical angles to solve problems. Students will need to focus on relevant details within a problem to know how to create the equations and solve them. Support students to develop generalizations based on the similarities found among problems.
	3.6.2 Activity provides a kinesthetic entry point using a Card Sort. Students will work with a partner or small group to match the name of the angle pair, image of the angle pair, and the value of the variable when solving using the angle pair relationship.	
Lesson Notes:		

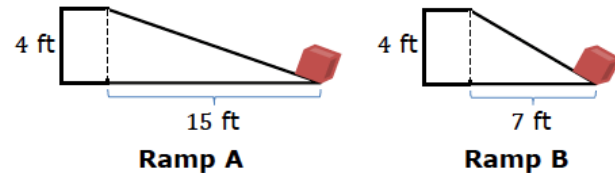
3.6.1 Warm-Up: Would You Rather?

Component Overview: Students choose whether they would rather push a box up of one of two ramps with different dimensions.



3.6.1 Warm-Up: Would You Rather?

- Two ramps set up to move boxes up to a 4 foot (ft) high platform are shown.



- Would you rather push a 50 pound (lb) box up Ramp A or Ramp B?

Answers will vary.

I would rather push my box up Ramp A because it is less steep.

I would rather push my box up Ramp B because the ramp is shorter.

- Justify your choice.

Answers will vary.

The slope of Ramp A is $\frac{4}{15}$ or approximately 0.27 making it less steep than Ramp B which has a slope of $\frac{4}{7}$ or approximately 0.57. I would rather push the box up a less steep incline.

The length of Ramp B is $\sqrt{65}$, or approximately 8, ft long while Ramp A is approximately 15 ft long. I would rather push the box a shorter distance.



Students may compare steepness or may compare length. Both are appropriate with the correct justification.



Do you know someone who is a carpenter, architect, or engineer? Consider inviting a guest speaker to join your class (virtually or in person). Students can ask questions to the guest speaker. If you have a Career Technical Education (CTE) program, have the instructor and some of their students come to your class to discuss this scenario.

3.6.2 Activity: Solving Using Angle Relationships, Postulates, and Theorems

Component Overview: Students will work with a partner or small group to complete a Card Sort. They will match the name of an image with the name of the angle relationship labeled and then use the information to find the matching value for the variable.



3.6.2 Activity: Solving Using Angle Relationships, Postulates, and Theorems

With your partner, sort the cards to match each image with the type of angle pair represented and the value of x . Record your answers in the table, showing your work in the last column.

Type of Angle Pair	Letter of Image	Work and Answer
Vertical Angles	A	$x + 25 = 2x + 5$ $20 = x$
Linear Pair	B	$(7x - 46) + (3x + 6) = 180$ $x = 22$
Alternate Interior Angles	C	$5x - 34 = 3x + 16$ $x = 25$
Alternate Exterior Angles	D	$13x + 9 = 14x + 2$ $7 = x$
Corresponding Angles	E	$3x - 33 = 2x + 26$ $x = 59$
Consecutive Exterior Angles	F	$(3x + 10) + (5x + 90) = 180$ $x = 10$
Consecutive Interior Angles	G	$(x + 15) + (3x + 5) = 180$ $x = 40$



Use observational data from the previous lessons to determine a small group of students who may benefit from additional support during this component. Consider pulling this small group of students to work with them through the activity before they practice on their own in the next component.



Group Presentations

After completing 3.6.2 Activity for each matching set, consider having a different pair of students present their answers to the class. Students should use proper academic language when describing the pairings and when justifying how to create the equations. After a pair finishes, have another student restate why the students made the matches they did.

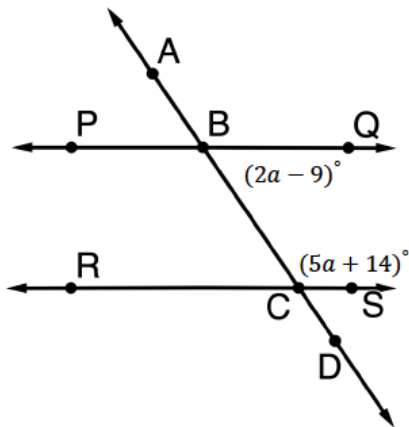
3.6.3 Your Turn!

Component Overview: Students apply what they have learned about various angle pair relationships to solve questions.



3.6.3 Your Turn!

1. $\overleftrightarrow{PQ} \parallel \overleftrightarrow{RS}$, where $\overleftrightarrow{AD}$ is a transversal.



- Determine the value of a .
25
- What is $m\angle ABQ$?
 139°
- What is $m\angle BCR$?
 41°



If students are struggling, consider using this time to work with small groups or one-on-one. Consider a peer review process for students to check each other's work as an added layer of support.

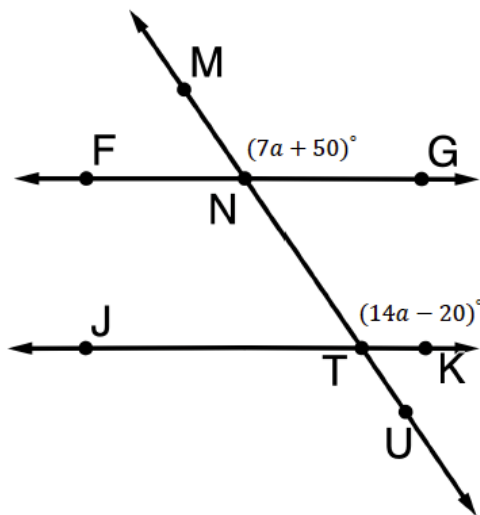


While monitoring students, consider asking the following probing questions, as needed:

- What is the relationship between the measures of these two angle pairs?
- What strategy will you use to solve this question?
- Why did you pick this strategy?

3.6.3 Your Turn! (continued)

2. $\overleftrightarrow{FG} \parallel \overleftrightarrow{JK}$ where $\overleftrightarrow{MU}$ is a transversal.



- a. Determine the value of a .

10

- b. What is $m\angle FNT$?

120°

- c. What is $m\angle KTU$?

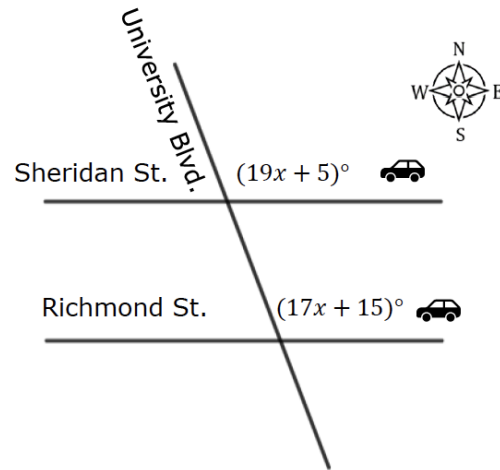
60°



For an added challenge, have students justify their answers by organizing their work in a two-column table with each step of their work on the left side, and the explanation on the right.

3.6.3 Your Turn! (continued)

3. Sheridan Street and Richmond Street are parallel, and both are intersected by University Boulevard.



Two cars are heading west, one on Sheridan St. and one on Richmond St. Both cars turn north onto University Blvd. What is the measure of the angle of their turn onto University Blvd.?

100°

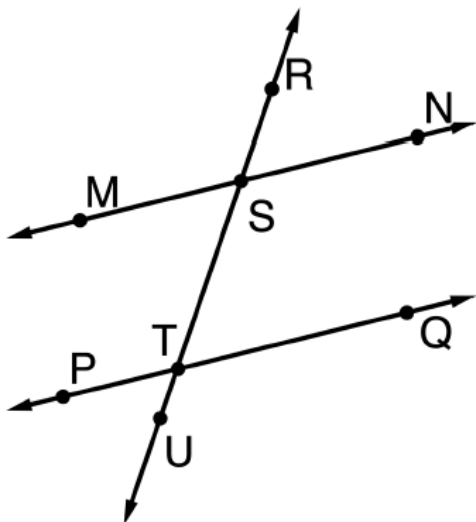
3.6.4 Wrap-Up: Ticket Out the Door

Component Overview: Students apply their knowledge of angle relationships, postulates, and theorems to find missing angle measures.



3.6.4 Wrap-Up: Ticket Out the Door

1. Given $\overleftrightarrow{MN} \parallel \overleftrightarrow{PQ}$, $m\angle PTS = (14x - 10)^\circ$, and $m\angle NST = (9x + 35)^\circ$, find each angle measure.



- a. $m\angle MSR$
 116°
- b. $m\angle STQ$
 64°



Use the data collected during this activity to develop next steps for student support. Consider the following data points:

- Did the student label the angles with the correct expression? If not, students may need remediation on naming angles.
- Did the student use the correct angle relationship to solve for the variable? If not, students may need remediation on identifying the angle pairs and the theorems associated with them.
- Did the student solve the equation correctly? If not, students may need remediation on solving multi-step equations.
- Did the student find the angle measure asked for? If not, students may need strategies for reading the question to determine what they are looking for when solving a problem.